

课程总览



什么叫 bug

- Software Fault : A static defect in the software (i.e., defect) 静态错误
- Software Error : An incorrect internal state that is the manifestation of some fault 状态错误
- Software Failure : External, incorrect behavior with respect to the requirements or other description of the expected behavior 输出行为错误



PIE 模型

1. Execution: The location or locations in the program that contain the fault must be **reached**
2. Infection : The **state** of the program must be incorrect
3. Propagation : The infected state must propagate to cause some **output** of the program to be incorrect



Bug 空间聚集

故障通常聚在一起。

测试的局限性

输入空降庞大：无法穷举所有输入

实现逻辑复杂：无法想到所有场景

测试语言未知：无法判定测试的预期输出

发现的 Bug 类型

- 断言失败
- 异常崩溃
- 无效输入
- 错误输出